

Comparing the results of R functions to Csnippets

Katie Montovan

2025-01-29

This document is focused on comparing the simulation results from a simple SI model setup in R functions to the same model implemented in Csnippets. It can be helpful to modify this file to sort out a new model when you are switching to csnippets, and then copy and paste the step, rinit, rmeasure, and dmeasure functions back into your main file once you get it working.

Load in data - from github, your computer, etc.

```
library(readr) #this library is used to help load the csv file.
simuldata <- read_csv("https://raw.githubusercontent.com/kmontovan/pomp/refs/heads/main/SimulatedPompiti")

## Rows: 150 Columns: 2
## -- Column specification -----
## Delimiter: ","
## dbl (2): day, NewCases
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

R functions

```
#Setup Model Step function
R_step <- function(S, I, NewI, Beta, gamma, delta.t, ...){
  infections <- rbinom(n=1,size=S, prob=1-exp(-Beta*I*delta.t))
  recover <- rbinom(n=1,size=I, prob=1-exp(-gamma*delta.t))
  S <- S - infections + recover
  I <- I + infections - recover
  NewI <- NewI + infections
  c(S=S, I=I, NewI=NewI)}

R_rinit <- function(N, phi, ...){
  c(S = round((1-phi)*N),
    I= round(phi*N), NewI=0)}

R_rmeas <- function(NewI, rho,k,...){
  c(NewCases=rnbinom(n=1,size=k,
                    mu=rho*NewI))}

R_dmeas <- function(log,NewCases, NewI, rho, k, ...){
  dnbinom(x=NewCases, size=k, mu=rho*NewI, log=log)
}
```

Csnippets

```
C_step <- Csnippet("
  double infections = rbinom(S,1-exp(-Beta*I*dt));
  double recoveries = rbinom(I,1-exp(-gamma*dt));
  S += -infections + recoveries;
  I += infections - recoveries;
  NewI += infections;
")

C_rinit <- Csnippet("
  S = nearbyint((1-phi)*N);
  I = nearbyint(phi*N);
  NewI = 0;
")

C_rinit <- Csnippet("
  S = N-4;
  I = 4;
  NewI = 0;
")

C_rmeas <- Csnippet("
  NewCases = rnbinom_mu(k,rho*NewI);
")

C_dmeas <- Csnippet("
  lik = dnbinom_mu(NewCases,k,rho*NewI, give_log);
")
```

Setup Pomp structure with the R function components

```
SI_ModelR <- pomp(
  data = simuldata,
  times="day",
  t0=0,
  statenames=c("S","I","NewI"),
  accumvars="NewI",
  paramnames=c("Beta","gamma","rho","k","N","phi"),
  params= c(N=1000, #population size
            phi = 0.004, #fraction of population infected at time 0
            Beta=.0024, # rate of spread of illness - per day
            gamma=.1, #recovery rate
            rho=.8, #fraction of new infections that are recorded
            k=5 ), #changes the variability of reporting accuracy
  rinit=R_rinit,
  rprocess=euler(step.fun = R_step,delta.t= 1/4),
  rmeasure=R_rmeas,
  dmeasure=R_dmeas
)
```

Setup Pomp structure with the R function components

```
SI_ModelC <- pomp(  
  data = simuldata,  
  times="day",  
  t0=0,  
  statenames=c("S","I","NewI"),  
  accumvars="NewI",  
  paramnames=c("Beta","gamma","rho","k","N","phi"),  
  params= c(N=1000, #population size  
            phi = 0.004, #fraction of population infected at time 0  
            Beta=.0024, # rate of spread of illness - per day  
            gamma=.1, #recovery rate  
            rho=.8, #fraction of new infections that are recorded  
            k=5 ), #changes the variability of reporting accuracy  
  rinit=C_rinit,  
  rprocess=euler(step.fun = C_step,delta.t= 1/4),  
  rmeasure=R_rmeas,  
  dmeasure=C_dmeas  
)
```

Compare

Simulate the results and plot them on the same graph for the R functions and Csnippet versions of the model to make sure they are doing roughly the same thing. If there is an error somewhere, they might be doing drastically different things, so this is just a quick check.

```
simsR <- SI_ModelR |>  
  simulate(  
    nsim=25,  
    format="data.frame",  
    include.data=FALSE  
  )  
  
simsC <- SI_ModelC |>  
  simulate(  
    nsim=25,  
    format="data.frame",  
    include.data=FALSE  
  )  
  
ggplot()+  
  geom_line(data=simsR, aes(x=day,y=NewCases,group=.id),color="blue",alpha=.5)+  
  geom_line(data=simsC, aes(x=day,y=NewCases,group=.id),color="gray",alpha=.5)+  
  guides(color="none")+  
  theme_bw()
```

